COMPUTATIONAL PROTEOMICS · 2026

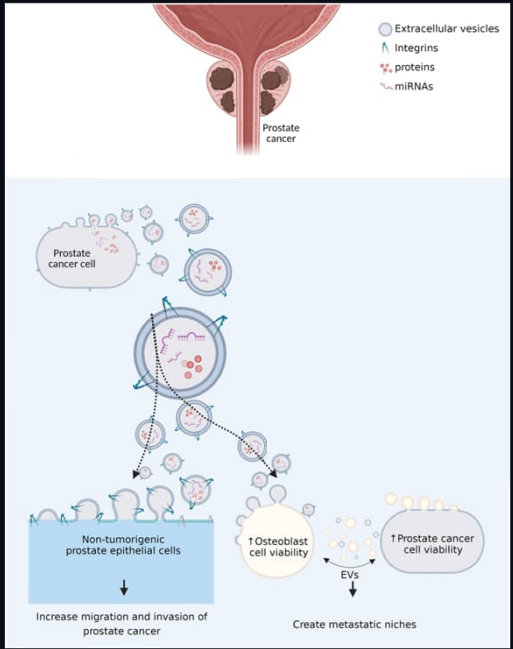
SILAC-TMT-RTS-SPS-MS3 Pipeline

Quantitative Proteomic Profiling of Differential Extracellular Vesicle Uptake in Osteoblastic Niches

Resolving Selective EV Internalization via Multiplexed Isobaric Labeling, Robust Scaling, and Orthogonal Machine Learning

Work done and presented by Vladislav Mantsevich

October 12, 2026



https://www.frontiersin.org/files/Articles/746040/xml-images/fonc-11-746040-g001.webp

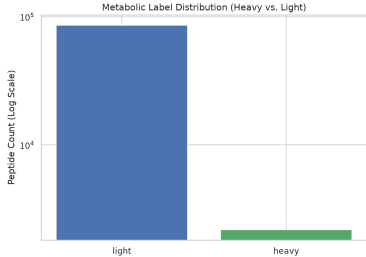
Discrimination of EV Cargo: SILAC-TMTpro 18-plex Workflow

- **Metabolic Cargo Tracing:** Donor cells labeled with heavy isotopes (Arg10 / Lys-D8 SILAC); unlabelled recipient SAOS-2 proteome serves as light background (97.5% of raw pool).
- **Multiplexed Cohort**: 18-plex TMTpro design including boost channels, empty negative controls, baseline SAOS-2 controls (N=3), uptake conditions (PC3 N=3, PC3M N=4), and pure EV spike-ins (N=2).
- **High-Purity Quantitation**: Orbitrap Eclipse running SPS-MS3 with MIPS precursor filtering and strict co-isolation threshold (<75%) to eliminate reporter ion compression.
- **Identification Rigor**: Dual-branch Sequest HT searches with Percolator SVM target-decoy scoring (q <= 0.01 at PSM and protein levels).

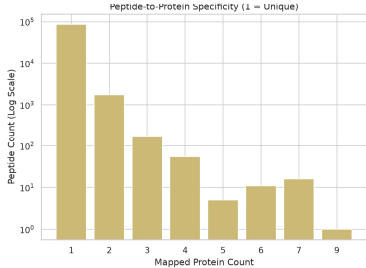
TMT channel	Sample content
126	PC3 Evs
127N	PC3M Evs
127C	
128N	
128C	SAOS2
129N	SAOS2
129C	SAOS2
130N	SAOS2 + PC3 EV
130C	SAOS2 + PC3 EV
131N	SAOS2 + PC3 EV
131C	SAOS2 + PC3M EV
132N	SAOS2 + PC3M EV
132C	SAOS2 + PC3M EV
133N	SAOS2 + spike PC3 EV
133C	SAOS2 + spike PC3 EV
134N	SAOS2 + spike PC3M EV
134C	SAOS2 + spike PC3M EV
135N	SAOS2 + PC3M EV

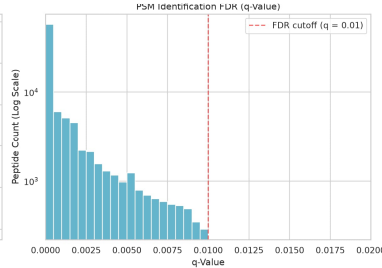
Upstream Quality Control (MS1 and MS2 Levels)

1. **Metabolic Label Ratio**: Proportion of Heavy to Light peptides.
2. **Contaminant Annotation**: Verification that external contaminants were filtered out.
3. **Peptide Uniqueness**: Frequency of peptide matches mapping to a distinct proteins.
4. **PSM False Discovery Rate**: Global q-value distribution to ensure high-confidence identification thresholds.



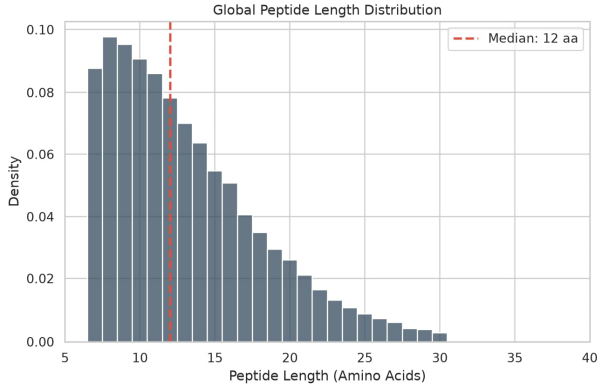


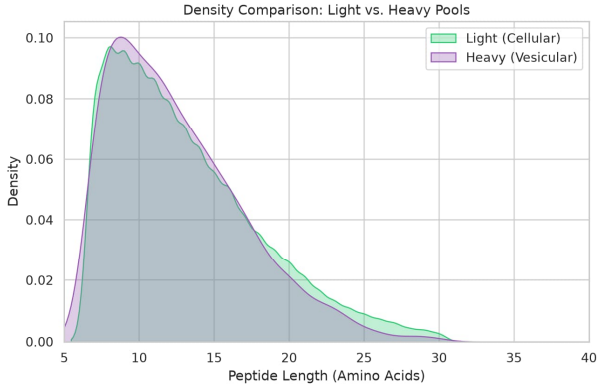




Enzymatic Cleavage and Peptide Length Profiling

```
Sequence length summary metrics:
Total peptides: 88138
Minimum length: 7 aa
Median length: 12.0 aa
Mean length: 13.13 aa
Maximum length: 30 aa
```

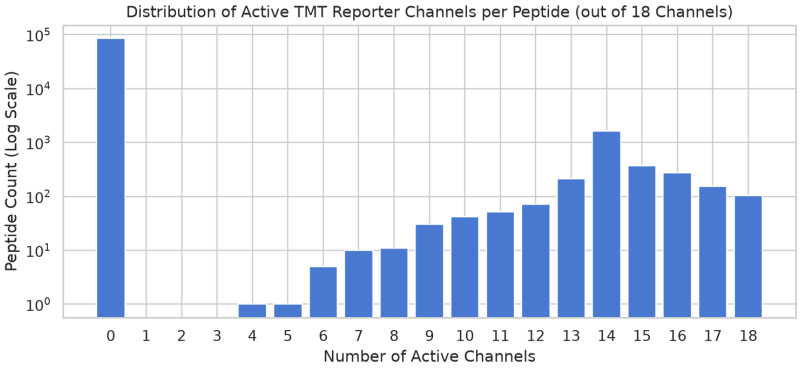




TMT Multiplex Reporter Occupancy Diagnostics

Channel summary by isotopic label:			
Label	count	median	mean
heavy	2209	0.0	6.63
light	85929	0.0	0.32

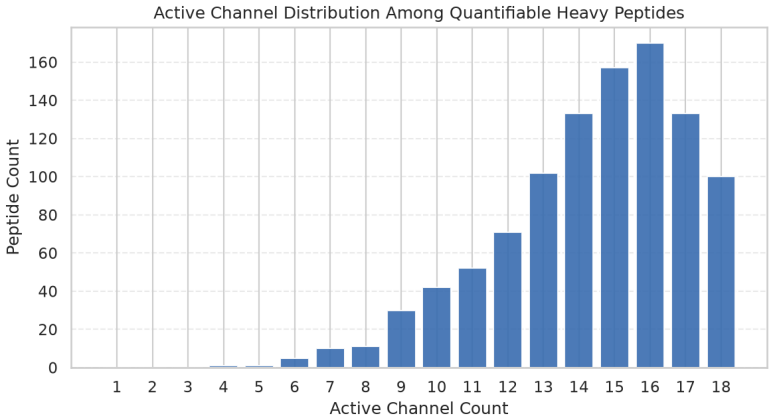
Within the target heavy subset (n=2,209), more than a half contain zero reporter intensities across all channels. Why?




```
Quantification status for Heavy peptides:
Quan Info
NoQuanValues        1175
NaN                 1018
ExcludedByMethod      16
Name: count, dtype: int64
```

Quantifiable Heavy Peptide Cohort

We isolate the 1,018 quantifiable heavy peptides (where are at least 1 non-zero channel) and examine their channel occupancy across replicates.



Protein Group Resolution and Shared Peptide Evaluation

All 1,018 peptides belong to a single protein group (Number of Protein Groups == 1).

However, 73 peptides map to multiple closely related proteins within the group (Number of Proteins > 1).

Strict exclusion of these 73 shared peptides would completely discard 20 proteins, including canonical EV cytoskeletal markers (ACTB, ACTC1, TUBA4A, VIM) whose tryptic fragments are identical between isoforms. Therefore, we track both cohorts: derived exclusively from 945 strictly unique peptides (373 proteins) (done), and derived from all 1,018 peptides (393 proteins) (in progress)

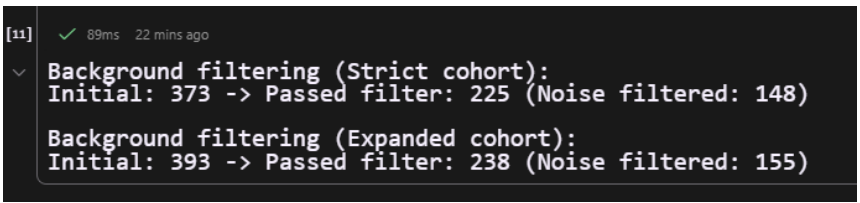
```
Proteins identified by strictly unique peptides: 373
Proteins identified exclusively via shared peptides: 20
Excluded candidate examples: O00571 O14493 O14818 P04908 P05388 P07910 P08670 P0C0S8
```

Background Noise Filtering Against Recipient Cell Controls

To eliminate non-specific background signals, a filter is being applied relative to the SAOS-2 alone control channels:

1. Peak sample median intensity must be higher than noise:
   Max[median(PC3 + EV), median(PC3M + EV)] > 5.0

1. Sample signal must be at least twofold above background:
   *Max[median(PC3 + EV), median(PC3M + EV)] >= 2 * median(SAOS2 control)*



Spike-in Normalization and Input Scaling

To account for EV input loading differences, sample intensities are normalized to the internal pure-EV spike channels.

A 2x NTA particle adjustment factor is applied to PC3 channels (−1.0 on the log2 scale):

$$\log_2(I_{norm}) = \log_2(I_{sample}) - \log_2(I_{spike}) - offset$$

Missing values are retained as *NaN* **without imputation** to prevent artificial variance reduction.

```
12]  ✓ 92ms  27 mins ago

Missing value count per protein across 7 channels:
0    159
1     63
2      3
Name: count, dtype: int64

Complete cases (quantified across all 7 channels): 159
```

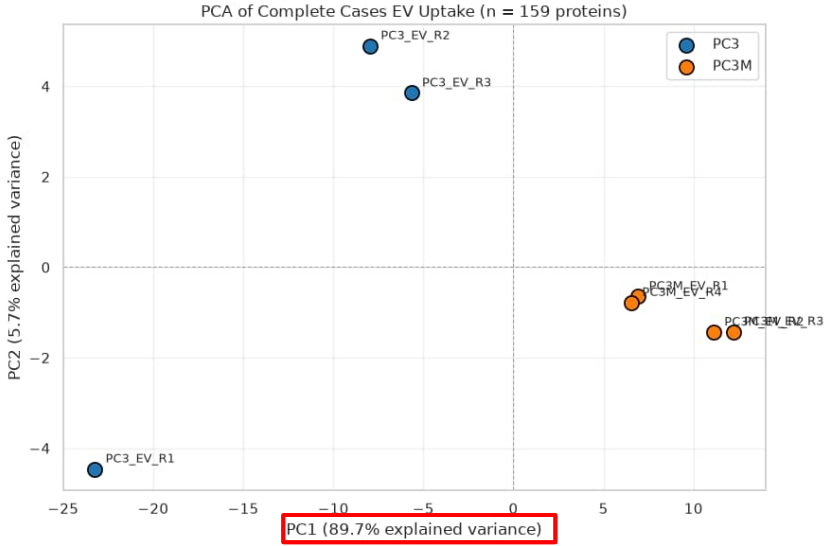
EV-cell incubation

To prevent excessive differences in quantities of isolated PC3 and PC3M EVs, their concentration was determined by NTA. Briefly, 1 µl of resuspended EVs was added to 999 µl PBS and measured on 3 different camera levels (12, 14, 16) for 3 times 30 seconds. The average of these measurements showed that there were ~2× more PC3 EVs than PC3M EVs. Since we did not want to risk adding a lower amount of PC3 EVs than with our initial EV*trace* experiments (REF), we decided to add an equal volume of the different EVs to the cells, even if this would mean that we added twice as much PC3 EVs than PC3M EVs. Thus in all cases, the amount of added EVs was derived from one T175 flask, which was equal to 30 mL of EV collection medium. The 58 µl of resuspended PC3 or PC3M EVs was mixed with 100 µl of McCoy 5a Glutamax medium (Gibco) supplemented with 2% FBS and pen/strep. Seeded SAOS2 cells in the 48 wells plate were first washed twice with 500 µl PBS before adding 150 µl of McCoy 5a + 2%

Commented [MK22]: Not sure if we should leave this in. Can also ignore, and just say we added XX volume of EVs to each well. Depends on how much emphasis we put in this difference along the results/analysis (since it messed up the normalization, right?)

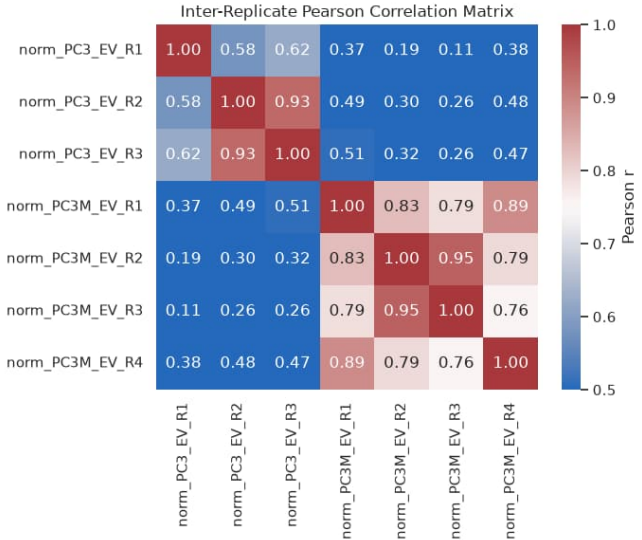
Exploratory PCA and Replicate Evaluation Policy

PCA is being performed across complete cases
(**n=159** – watch previous slide!)



Exploratory PCA and Replicate Evaluation Policy

Pearson correlation is being computed across complete cases
(**n=159** – watch previous slide!)



Identification of the Global Scale Effect

All the PC1 feature coefficients are positive!!! (+0.018 to +0.083)

In uncentered PCA, a component with uniform positive signs reflects a **global scale factor**

Meaning, recipient SAOS-2 cells internalize metastatic PC3M cargo at a systematically higher absolute volume than PC3 cargo.

✓ 5.7s Just now

Top 10 PC1 Loadings (Global Scale Dominance):

Master Protein Accessions	PC1_Loading	Global_Log2FC_PC3M_vs_PC3
P23229	0.083585	3.313838
P21589	0.083554	3.535186
P35613	0.083526	3.252563
P09543	0.083452	3.596595
O00468	0.083433	3.660825
P51149	0.083427	2.099352
Q9NQ84	0.083392	2.657023
P17301	0.083383	2.604376
Q06830	0.083378	2.200017
Q07020	0.083366	2.420473

Anti-Top 10 PC1 Loadings:

Master Protein Accessions	PC1_Loading	Global_Log2FC_PC3M_vs_PC3
P07900	0.018266	0.339007
Q9H4M9	0.042299	2.798268
Q12852	0.047613	0.221920
P62995	0.059084	1.610589
P49006	0.061684	1.521139
P29966	0.065440	2.641612
Q8TCT9	0.065774	0.579076
Q12907	0.066214	0.929701
P30050	0.066503	0.732596
P49207	0.067400	2.440638

Sample-Median Centering: Isolating Relative Cargo Composition

To isolate specific cargo selectivity from bulk endocytosis rates, we subtract sample medians:

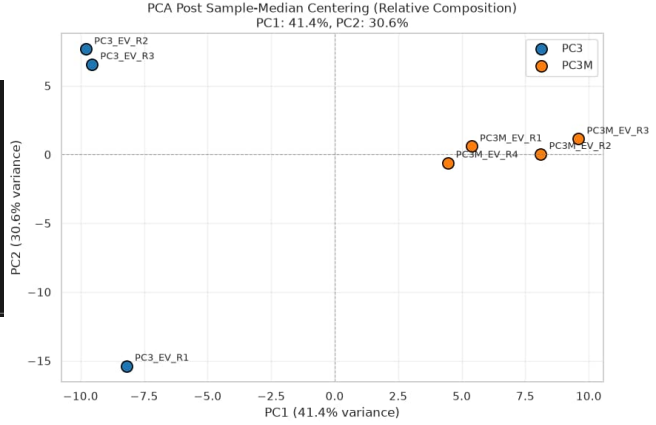
$$Signal_{rel;\,peptide,sample} = Signal_{peptide,sample} - Median(Signal_{peptide,sample})_{sample}$$

Top Proteins with Relative Cargo Selectivity (PC3M enriched):

Master Protein Accessions	PC1_Loading_Centered	Relative_Delta_PC3M_vs_PC3
P08572	0.122631	5.819160
P02751	0.119917	3.997408
P21980	0.120602	3.822650
P56199	0.108836	2.470602
P24821	0.117487	2.212438

Top Proteins with Relative Cargo Selectivity (PC3 enriched):

Master Protein Accessions	PC1_Loading_Centered	Relative_Delta_PC3M_vs_PC3
Q12852	-0.121389	-2.101391
P07900	-0.088742	-1.984304
Q8TCT9	-0.120919	-1.744235
P30050	-0.111290	-1.590715
P60903	-0.099464	-1.418956

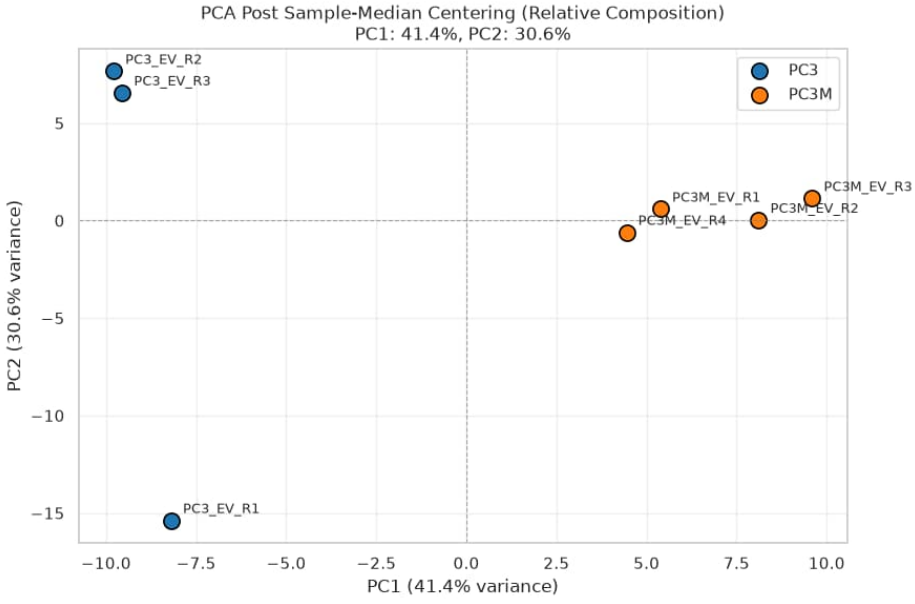


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P60903	-0.099464	-1.418956


PCA Post Sample-Median Centering (Relative Composition)
PC1: 41.4%, PC2: 30.6%
PC3
PC3M
PC3_EV_R2
PC3_EV_R3
PC3_EV_R1
PC3M_EV_R1
PC3M_EV_R2
PC3M_EV_R3
PC3M_EV_R4
PC2 (30.6% variance)
PC1 (41.4% variance)
5
0
−5
−10
−15
−10.0
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0

Univariate Differential Uptake Analysis: Welch's t-Test

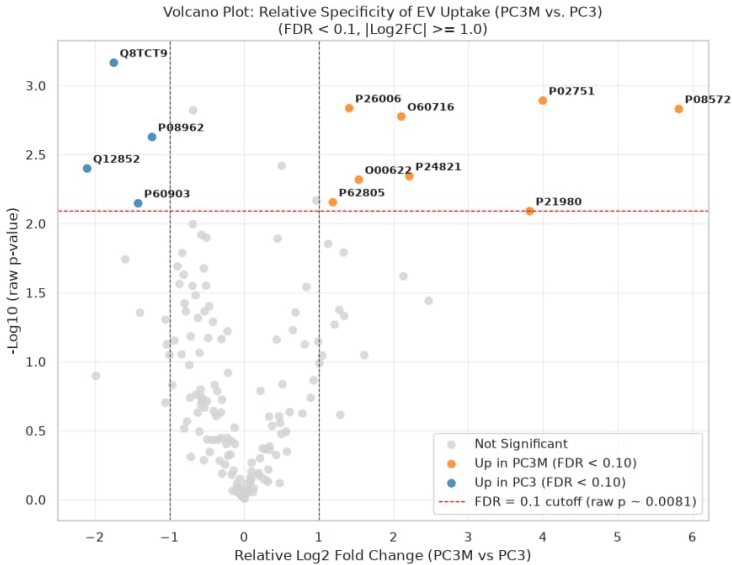
1. Heteroscedasticity:
   MS signal variance
   scales with intensity

2. Unbalanced Groups:
   n_1 = 3, n_2 = 4

Student's t-test assumptions are not met, but its variation – (two-sided) Welch's t-test – would work here well

Multiple testing correction is applied using the Benjamini–Hochberg False Discovery Rate:

FDR < 0.10,
log_2(FC) >= 1.0



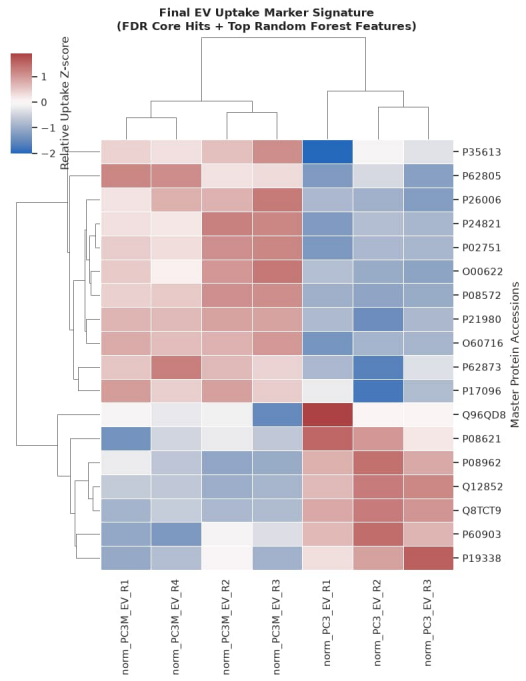
Univariate Differential Uptake Analysis: Welch's t-Test

Proteins meeting FDR < 0.1 and |Log2FC| >= 1.0: 12

Accession	Log2FC	p_value	q_value	Significance
O60716	2.104834	0.001675	0.044389	Up in PC3M (FDR < 0.10)
P02751	3.997408	0.001285	0.044389	Up in PC3M (FDR < 0.10)
P08572	5.819160	0.001479	0.044389	Up in PC3M (FDR < 0.10)
P26006	1.406785	0.001460	0.044389	Up in PC3M (FDR < 0.10)
Q8TCT9	-1.744235	0.000683	0.044389	Up in PC3 (FDR < 0.10)
P08962	-1.233099	0.002358	0.053565	Up in PC3 (FDR < 0.10)
P24821	2.212438	0.004535	0.069467	Up in PC3M (FDR < 0.10)
O00622	1.535682	0.004806	0.069467	Up in PC3M (FDR < 0.10)
Q12852	-2.101391	0.003987	0.069467	Up in PC3 (FDR < 0.10)
P62805	1.187132	0.007002	0.080947	Up in PC3M (FDR < 0.10)
P60903	-1.418956	0.007127	0.080947	Up in PC3 (FDR < 0.10)
P21980	3.822650	0.008123	0.086109	Up in PC3M (FDR < 0.10)

Hierarchical Clustering

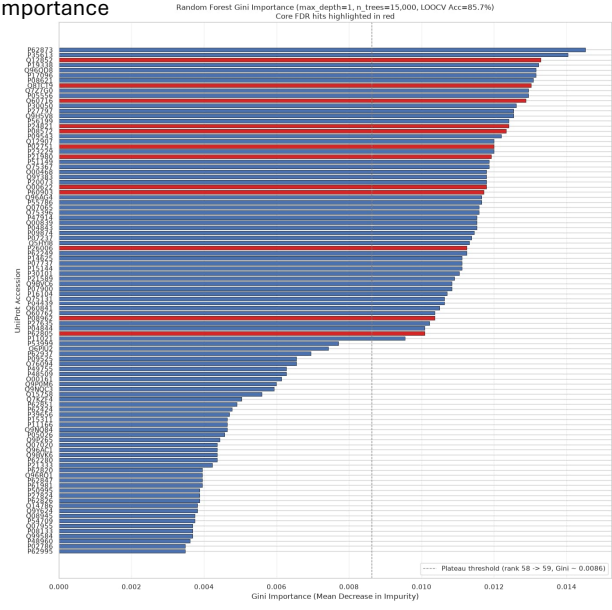
Lets take a look how well those statisticly important proteins separate conditions:

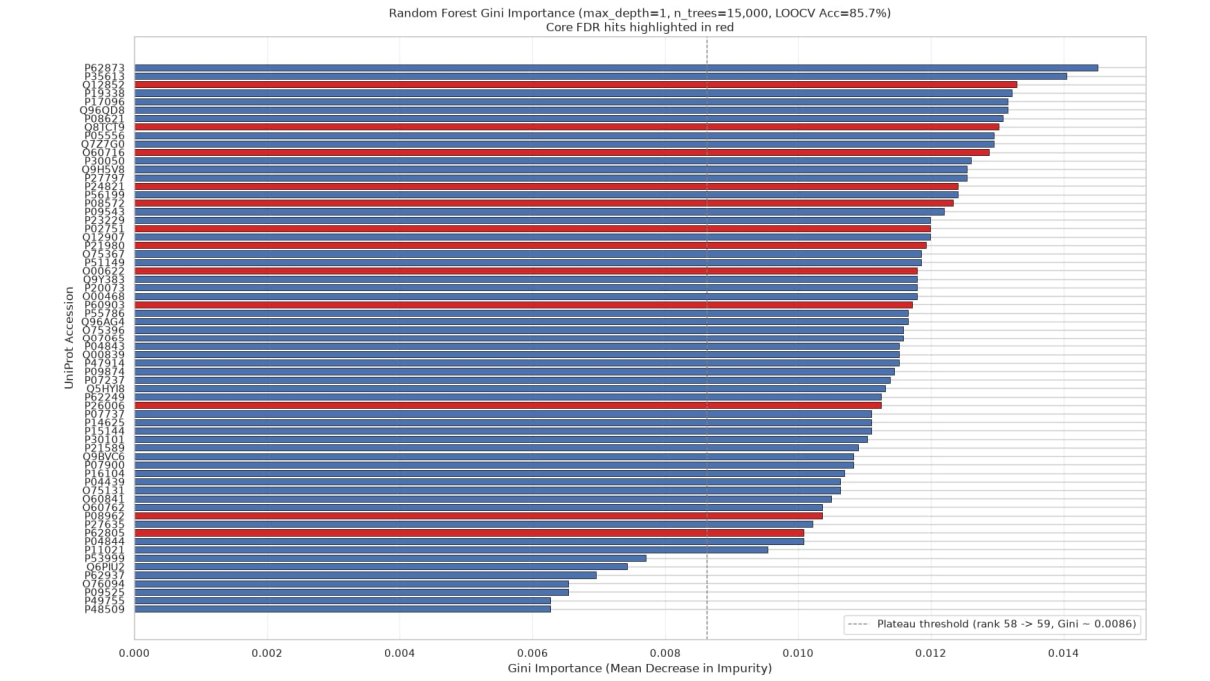


Supervised Machine Learning: Random Forest Feature Importance

To complement hypothesis testing with a non-parametric multivariate model, we train a Random Forest classifier: *max_depth=1; n_estimators=15000 trees*

Generalization accuracy is audited via Leave-One-Out Cross-Validation (LOOCV).




Random Forest Gini Importance (max_depth=1, n_trees=15,000, LOOCV Acc=85.7%)
Core FDR hits highlighted in red
UniProt Accession
P62873
P35613
Q12962
P19338
P17096
Q96QD8
P08621
Q8TCT9
P05556
Q7Z7G0
Q60716
P30050
Q9H5V8
P27797
P24821
P56199
P08572
P09543
P23229
P02751
Q12907
P21980
Q75369
P51149
O00622
Q9Y383
P20073
O00468
P60903
P55786
Q96AC4
Q75390
Q07065
P04843
Q00839
P47914
P09874
P07237
Q5HYI8
P62249
P26006
P07737
P14625
P15144
P30101
P21589
Q9BVC6
P07900
P16104
P04439
Q75131
Q60841
Q60762
P08962
P27635
P62805
P04844
P11021
P53999
Q6PIU2
P62277
Q76094
P09525
P46755
P48509
0.000
0.002
0.004
0.006
0.008
0.010
0.012
0.014
Gini Importance (Mean Decrease in Impurity)
Plateau threshold (rank 58 -> 59, Gini ~ 0.0086)

To examine the agreement between univariate hypothesis testing and multivariate machine learning, we project Random Forest importance ranks onto the Volcano plot.

A continuous color and point-size gradient highlights proteins within the top-58 importance plateau that fall just below stringent FDR thresholds due to degrees-of-freedom penalties.

